

TAM THUC STRATEGEM

DevOps, CI/CD and Infrastructure v1.0

1. CI/CD PIPELINE

- **GitHub Actions** workflow:
 - Lint + Type check
 - Unit + Integration tests
 - Build Docker images
 - Security scan (Trivy / Snyk)
 - Deploy to Staging → Production (with approval gate)
- **Infrastructure as Code**: Terraform + Kubernetes manifests.

2. INFRASTRUCTURE

- **Cloud**: AWS or GCP (multi-region for latency).
- **Kubernetes** cluster with auto-scaling.
- **Database**: Managed PostgreSQL + Read replicas.
- **Vector DB**: Pinecone or self-hosted Weaviate.
- **Monitoring**: Prometheus + Grafana + Alertmanager.
- **Logging**: Loki + Grafana.
- **Backup**: Daily automated backups + point-in-time recovery.

3. DISASTER RECOVERY

- RPO: 1 hour
- RTO: 4 hours
- Multi-AZ deployment + cross-region failover plan.